Interlink MVP – Android App Specification

For the **initial MVP of the Android application**, the primary goal is to help users manage their calendar events intelligently by syncing with Google Calendar, estimating travel time, and notifying the user ahead of time so they can decide whether they can attend the event.

Below is the proposed workflow and feature set.

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1. Google Calendar Synchronization

When a user first signs into the app, they will be required to **connect and sync their Google Calendar**.

The system will then:

- Import all existing events from the user's Google Calendar.
- Store these events inside the application.
- Automatically keep the app synchronized with Google Calendar in the future so that any **newly created or updated events** appear in the app without requiring manual syncing.

This ensures the user does not need to manually update events within the application.

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2. Chronological Event Listing

All synced events will be displayed inside the app in a **chronological list**.

Each event entry will include:

- Event title
- Event date and time
- Event location
- Organizer information
- Attendees (if available)

This list will act as the user's **central event dashboard**.

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3. Travel Time Estimation

For each event that has a location attached:

- The app will use the **user's current location**.
- It will query **Google Maps APIs** to determine:
  - Distance to the event location
  - Estimated travel time

This estimated travel time will be used to determine **when the user should be notified before the event starts**.

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4. Smart Notification Timing

Notifications will be scheduled based on a calculated buffer time.

The notification time will be determined using the following formula:

```
Notification Time =
Event Start Time
− (Estimated Travel Time + Preparation Buffer Time)
```

Example:

- Event Start Time: **5:00 PM**
- Estimated Travel Time: **30 minutes**
- Preparation Buffer: **15 minutes**

```
Notification Time = 4:15 PM
```

This ensures the user has enough time to:

- prepare
- leave
- reach the event on time.

For the MVP, the **preparation buffer time will be fixed**, but later it can be made customizable.

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5. Event Attendance Prompt

When the notification triggers, the user will see a **popup-style prompt** (similar to Google authentication popups).

This prompt will appear **whether the user is inside the app or outside the app**.

The notification will contain basic event details and, when tapped, will **deep-link the user to a specific screen inside the app**.

On this screen the user will be asked:

"Will you be able to attend this event?"

The user will have two options:

- **Yes, I'll attend**
- **No, I won't**

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6. If the User Selects "Yes"

If the user confirms attendance:

- The workflow ends.
- No further actions are required.

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7. If the User Selects "No"

If the user declines the event, the system will assist them in notifying the event participants.

The app will retrieve the following information from the calendar event:

- Organizer email address
- Attendee email addresses

An email will then be prepared for the user.

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8. Email Response Options

The app will provide **two email options**:

Option 1 — Default Email Draft

A predefined message will automatically be generated, such as:

> "Unfortunately, I will not be able to attend this event."

The user can send this message immediately.

Option 2 — Custom Email Draft

The user can choose to **write their own message**.

The app will allow the user to:

- Edit the message
- Update the message
- Delete and rewrite the message
- Send the email once ready

The email will be sent to:

- Event organizer
- All event attendees

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10. Automatic Future Sync

After the first synchronization:

- The app will **automatically keep the user's calendar events updated**.
- Any new events added to Google Calendar will automatically appear in the app.
- The user will not need to manually sync again.

Summary of MVP Features

The MVP version of the app will include the following core capabilities:

1. Sync Google Calendar events into the app.
2. Display all events in chronological order.
3. Estimate travel time to event locations using Google Maps.
4. Send smart notifications based on travel time + preparation buffer.
5. Show a popup asking whether the user can attend the event.
6. If the user declines, allow sending an email to the organizer and attendees.
7. Provide both default and customizable email drafts.
8. Automatically keep calendar events synchronized in the background.
